

Enhancing Digital Education Using VR Technology

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Abstract

Abstract: The context defines the shift toward immersive digital learning tools in higher education to replace passive 2D instruction. The brief literature summarises current Extended Reality (XR) literature, highlighting strong immediate engagement but limited long-term evaluation. The problem statement states the central problem regarding single-institution pilot sample sizes and the lack of theoretical knowledge retention data. The aim introduces the Scalable, Evidence-Based XR Integration Framework (SEB-XRIF), designed to track engagement and retention. The methodology states the adoption of the Design Science Research Methodology (DSRM). The dataset explains the use of the xAPI Educational Mining Dataset (480 records, 17 attributes) acquired from Kaggle. The methods specify the machine learning model selected (Random Forest Classifier) for student performance tiering. The evaluation metrics list the performance metrics utilised, including F1-score, System Usability Scale (SUS), and Cohen's d effect size. Anticipated results state the expected outcomes regarding predictive classification accuracy and long-term knowledge decay tracking. The research contribution and impact outline the broader contribution of providing higher education institutions with a scalable, evidence-based XR adoption framework.

Keywords: Extended Reality, Immersive Learning, Educational Data Mining, Design Science Research, Learning Analytics, Reproducibility.

1. Introduction

This section is written as a skeleton (context, literature, problem, objectives) because the full introduction is finalised once results are available. It is intentionally explicit about what each paragraph will cover.

Context and background. Establish the current landscape of digital education in applied fields such as Information Technology at the Durban University of Technology, focusing on the reliance on traditional 2D materials.

Literature and gaps. Review current XR adoption in higher education, identifying key limitations such as system latency, small single-institution sample sizes, and unverified theoretical knowledge retention.

Problem and motivation. Justify the need for a standardised, multi-attribute framework that connects real-time student engagement with long-term retention tracking.

Proposed solution. This study proposes the Scalable, Evidence-Based XR Integration Framework (SEB-XRIF). It addresses the retention and scalability gaps above, adopts the Design Science Research Methodology (DSRM), uses the public xAPI Educational Mining Dataset (480 records, 17 attributes) acquired from Kaggle, builds on a Python, Flask, React.js, and Scikit-learn stack with a Random Forest Classifier, and evaluates with F1-score, the System Usability Scale, and Cohen's d effect size.

1.2 Objectives

1. Review existing immersive pedagogical methods to pinpoint structural barriers preventing effective theoretical knowledge transfer.
2. Define a standardised framework architecture for XR assessment that accommodates longitudinal data gathering and multi-institutional adoption.
3. Build an interactive analytics dashboard prototype using the xAPI dataset to display participation trends and forecast student performance.
4. Assess framework usability and efficacy using the System Usability Scale (SUS) and Cohen's d effect size calculations.

2. Literature Review

The group systematic literature review covers 28 peer-reviewed studies from 2023 to 2026, selected under PRISMA 2020. It answers four questions on methods, data, metrics, and limitations. This section condenses it for the full

paper: existing approaches, theoretical evaluation with metrics and synthesis, evaluation of intelligent models, and technical considerations that motivate SEB-XRIF.

2.1 Existing Approaches

Virtual Reality dominates the evidence base with 16 of 28 studies, followed by Mixed Reality with 4 and Augmented Reality with 3. VR pairs with established instructional models, including a hybrid ADDIE and Borg and Gall design in the Preinanku language game and User-Centred Design with Problem-Based Learning in moot court simulations. AR supports handwashing training for children with autism spectrum disorder, while MR provides layered 3D pathology models for spatial reasoning.

Scale varies from 5,000 nursing students across 50 universities to controlled cohorts of 5, with a median of 30. Effectiveness concentrates on practical skills with gains above 30 percent and engagement near 92 percent, while theoretical transfer remains weak. Small samples and technical faults such as rendering latency in AI interview systems recur across the set.

2.2 Theoretical Evaluation and Synthesis

Evaluation follows a triad of learning outcomes at 85 percent, engagement at 77 percent, and usability at 58 percent. Representative instruments include effect sizes such as Cohen’s d of 0.936 and sentiment F1 of 91.7 percent from DistilBERT.

Synthesis selects VR simulation as the primary method because it carries the practical skill gains, with AR and MR retained as complements for overlays and spatial models. Microteaching specificity scores and museum usability dimensions show that measurement is maturing, but the absence of longitudinal retention data leaves the central contradiction unresolved: strong immediate gains against unknown durability.

2.3 Theoretical Evaluation of Intelligent Models

Only two reviewed studies apply intelligent models to learner data. DistilBERT reaches an F1 score of 91.7 percent on feedback sentiment, and regression on intent to use reaches R squared of 0.883.

The review provides no comparison of Random Forest, Support Vector Machine, or K-Nearest Neighbours, so these are not presented as SLR findings. Random Forest is selected prospectively for SEB-XRIF on external ensemble benchmarks for the xAPI dataset and its robustness to noisy educational data with ranked feature importance. Voice-driven dialogue remains fragile across accents and dialects, and abstract visualisation learning shows the same boundary between demonstration and evidence.

2.4 Technical Considerations

Delivery depends on headsets such as Oculus Quest, HTC Vive, and HoloLens 2, built with Unity and Blender content and speech-to-text or conversational AI. Evidence comes from questionnaires, feedback transcripts, and performance assessments.

The SLR does not consistently report programming languages, databases, or development environments, so these are not assumed. The specified foundation is therefore XR environments with 3D content, AI language technologies for feedback, and logged performance data, which Section 3 implements as Flask, React, and Chart.js over the xAPI schema.

3. Research Methodology

This study adopts the Design Science Research Methodology (DSRM) to guide the development and evaluation of the Scalable, Evidence-Based XR Integration Framework (SEB-XRIF). DSRM originates in information systems design research and was formalised by Peffers et al., and is applied here in five steps. It links each design decision to a documented problem, as in comparable immersive education work. The problem phase uses the group systematic literature review of 28 studies, which found retention rarely measured and a median sample of 30, alongside passive two-dimensional instruction at the Durban University of Technology.

3.1 Proposed Solution

SEB-XRIF is proposed with all choices derived from the SWOT analysis. The method is immersive virtual reality to address passive instruction. The dataset is the public xAPI Educational Mining Dataset for reproducibility and scale. The metrics are accuracy and F1 for the model, the System Usability Scale for the dashboard, and Cohen’s d for learning effect. The limitations addressed are small samples, absent retention measurement, and absent prediction on logged behaviour. The four layers are data, analytics, visualisation, and evaluation.

The framework pursues four objectives: (1) identify structural barriers to theoretical knowledge transfer; (2) define a standard XR architecture for longitudinal multi-institutional use; (3) build a dashboard prototype that visualises engagement and predicts performance; and (4) assess usability and effectiveness using the System Usability Scale and Cohen’s d.

3.2 Data Gathering

The study uses the xAPI Educational Mining Dataset from Kaggle by Amrieh et al. under Creative Commons Attribution ShareAlike 4.0. The structured table holds 480 Kalboard 360 records collected through Experience API, labelled Low, Medium, or High, with 16 predictors covering demographics, academic context, and behaviour. Counts are 127 Low, 211 Medium, and 142 High. The authors published it preprocessed, a completeness check found no missing entries.

A public dataset addresses the single-cohort median of 30 and supports reproduction and extension. The records are primary and secondary school level and therefore prototype the analytics layer only.

3.3 Design and Development

Design plans four layers with a pipeline from raw data through preprocessing, stratified split, and Random Forest to predicted tiers. Preprocessing applies one-hot encoding to categorical variables and retains behavioural counts, as tree models do not require scaling.

Tool selection follows the literature review theoretical evaluation, which ranked Random Forest strongest. The ensemble resists noise and overfitting and ranks feature importance, expected to highlight raised hands and visited resources. Configuration is 100 trees, balanced weights, and a fixed seed, with stratified 80 percent training and 20 percent testing. The demonstration dashboard uses Flask, React, and Chart.js to show trends, tiers, and importance.

3.4 Evaluation Criteria

Evaluation uses accuracy, per-class precision, recall, F1, and ten-fold stratified cross-validation, with expected accuracy of 75 to 83 percent from published benchmarks. Usability uses the System Usability Scale against 76.6 reported for extended reality training. Effectiveness uses Cohen’s d with thresholds 0.2, 0.5, and 0.8 against 0.936 for immersive practical training.

Results are compared against these baselines, and a Durban University of Technology pilot with one-semester follow-up supplies the missing longitudinal retention measure.

4. Technology Description

This section sells the chosen technologies to a policymaker: what each technology can do, why it matters for the country, what this study uses it for, and where its limits lie. The study builds no new headset or engine. It reuses established XR hardware with an analytics and evaluation framework on standard learning data.

Technology	Capability, national value, and study use
Virtual reality simulation	Brings safe practice without real-world consequences. Practical gains exceed 30 percent with engagement near 92 percent. Value is hands-on training at scale, as shown by 5,000 nursing students across 50 universities. The study uses VR as the instructional method for procedural skills, not for abstract theory where it remains weak.
Random Forest with language feedback	An ensemble of decision trees that tolerates noisy education data and ranks the behaviours that drive results. Expected accuracy is 75 to 83 percent on the xAPI benchmark, while DistilBERT scores feedback sentiment at 91.7 percent F1. Value is early warning for support, not gatekeeping. The study uses it to predict Low, Medium, or High tiers with explanations.
xAPI logging with dashboard	Experience API gives every institution the same standard event record. A Flask, React, and Chart.js dashboard shows trends, tiers, and importance to instructors. Value is comparability: any adopter

Technology	Capability, national value, and study use
	reports accuracy with ten-fold validation, usability against 76.6, and learning effect with Cohen's d, plus one-semester follow-up.

Limits are stated openly. Headset cost and dependence remain, with latency and voice recognition errors documented. Usability concerns persist across dimensions. The analytics prototype uses primary and secondary school records and therefore cannot claim higher education outcomes until the Durban University of Technology pilot with delayed testing is complete.

5. Developments

This section reports how the analytics pipeline was actually implemented. It is completed in T20, drawing directly from `reports/` and the MLflow runs so the numbers cannot drift from the code.

5.1 Data engineering

The raw xAPI Educational Mining Dataset is validated against a schema, versioned with DVC, and ingested into the framework. The implementation adds a real Experience API path: the records are represented as xAPI statements and loaded into a Learning Record Store, then read back into the analytics layer.

5.2 Analytics and the comparison matrix

A single scikit-learn pipeline performs one-hot encoding, a stratified split, and model fitting. The full comparison matrix spans sixteen estimators across classical, tree, boosting, kernel, instance-based, neural, and ensemble families, each evaluated with stratified ten-fold cross-validation.

5.3 Service and dashboard

The promoted pipeline is served through a FastAPI service exposing prediction, metrics, importance, and trend endpoints, and a React dashboard built on a shared component library visualises tiers, behaviour, and feature importance.

TODO (T20): replace the outline above with the implementation narrative, configuration, and reproducibility description.

6. Results

The comparison matrix trains sixteen estimators on the xAPI Educational Mining Dataset (n=480) with a held-out 80/20 split and stratified ten-fold cross-validation. Random Forest is the strongest model by cross-validated macro F1 (0.807), closely followed by stacking and extra trees. All tree-based and boosting families cluster within or above the published 0.75–0.83 accuracy benchmark for this dataset, while the dummy baseline confirms the task is not trivially learnable.

The table below reports accuracy, macro F1, cross-validated macro F1 with its standard deviation, one-vs-rest ROC-AUC, and fit time for every model. Stratified 10-fold cross-validation and a held-out 80/20 split on the xAPI Educational Mining Dataset (n=480).

Model	Accuracy	Macro F1	CV macro F1	CV std	ROC-AUC	Fit (s)
random_forest	0.7604	0.7688	0.8070	0.0493	0.9113	0.16
stacking	0.7708	0.7766	0.8002	0.0389	0.9197	1.86
extra_trees	0.7812	0.7857	0.7972	0.0535	0.9221	0.15
catboost	0.7812	0.7895	0.7924	0.0605	0.9022	1.41
mlp	0.7604	0.7676	0.7904	0.0347	0.9104	0.77
voting	0.8021	0.8074	0.7817	0.0699	0.9042	0.24
lightgbm	0.7500	0.7574	0.7815	0.0464	0.8956	0.25
xgboost	0.7292	0.7350	0.7789	0.0593	0.8900	0.23
gradient_boosting	0.7708	0.7770	0.7725	0.0552	0.8884	0.42
hist_gradient_boosting	0.7708	0.7747	0.7688	0.0513	0.8926	0.45
svc	0.7812	0.7857	0.7650	0.0723	0.8892	0.05

Model	Accuracy	Macro F1	CV macro F1	CV std	ROC-AUC	Fit (s)
logistic_regression	0.7500	0.7570	0.7641	0.0787	0.8874	0.03
decision_tree	0.6354	0.6446	0.7460	0.0820	0.7372	0.01
knn	0.7188	0.7276	0.7139	0.0712	0.8646	0.01
naive_bayes	0.6042	0.5970	0.5591	0.0932	0.8090	0.01
dummy	0.4375	0.2029	0.2036	0.0020	0.5000	0.01

Published ensemble benchmark for this dataset: 0.75–0.83 accuracy (Amrieh et al., 2016).

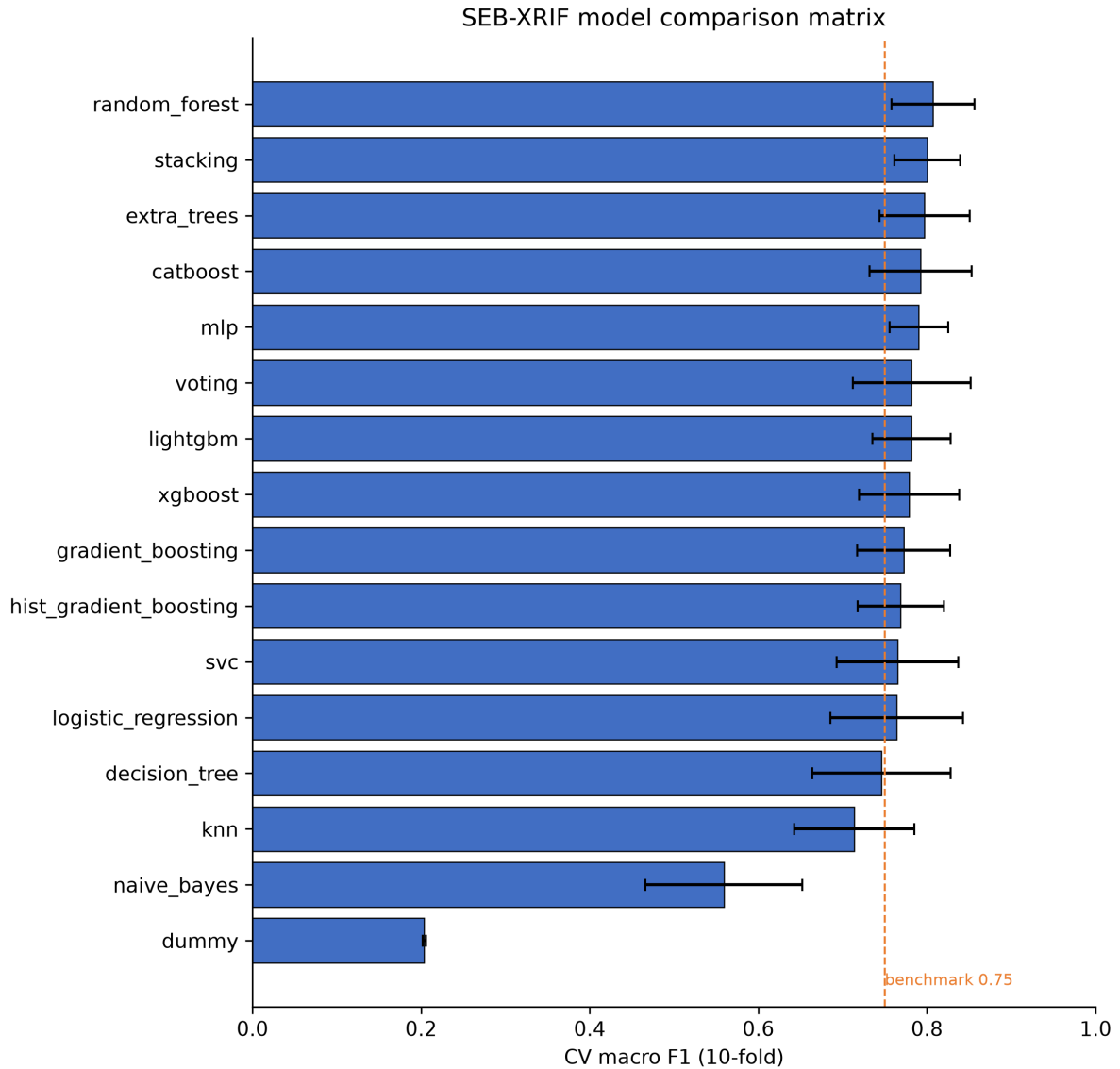


Figure 1: Model comparison matrix across the 16 estimators (CV macro F1 $\pm$ std).

7. Business Benefits

TODO (T20). This section will set out the value of SEB-XRIF for institutions and policymakers: comparable evidence across adopters, reusable xAPI data, reduced duplication of pilot effort, and an early-warning analytics capability that supports learners rather than gatekeeping them.

8. Conclusions

TODO (T20). This section will summarise the achievements (a reproducible, evidence-based XR integration framework with an analytics layer, service, dashboard, and fixed evaluation protocol), the further work needed (the one-semester delayed retention measurement and a higher-education pilot), and recommendations for adopters.

Acknowledgements

The authors would like to thank Durban University of Technology for providing the research environment and resource support that made this investigation possible. Special appreciation is extended to our lead supervisor and lecturer, Ms. Lindelweyizizwe Manqeke, as well as our mentor Kwanele Mazibuko for their guidance. We are also grateful to our co-authors and external advisor for their input, and to the researchers whose work formed the foundation of this study.

Declaration on the use of AI

The authors acknowledge the use of Artificial Intelligence tools during the preparation of this paper, specifically to assist with literature discovery and the maintenance of the reproducible analytics pipeline. The research team remained responsible for all study design, data analysis, and interpretation of results, and maintains full accountability for the integrity and originality of the final manuscript.

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